

LOFTUS, United Kingdom

WMO: 032750

Lat: 54.5628N

Lon: 0.8636W

Elev: 518

StdP: 14.42

Time Zone: 0.00 (EUW)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 28.0	(c) 29.9	(d) 21.8	(e) 16.7	(f) 30.4	(g) 24.1	(h) 18.6	(i) 32.2	(j) 44.7	(k) 46.0	(l) 41.3	(m) 45.4	(n) 10.5	(o) 250	(p) 0.744

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.7	(c) 73.6	(d) 64.0	(e) 70.5	(f) 62.1	(g) 67.8	(h) 60.5	(i) 65.4	(j) 71.2	(k) 63.5	(l) 68.6	(m) 61.8	(n) 66.2	(o) 12.6	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.1	88.2	67.8	61.3	82.5	65.9	59.8	78.2	64.1	30.6	71.6	29.1	68.7	27.9	66.2	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 35.7	(b) 31.3	(c) 27.8	(e) 24.5	(f) 80.1	(g) 2.9	(h) 3.4	(i) 22.4	(j) 82.5	(k) 20.8	(l) 84.4	(m) 19.2	(n) 86.3	(o) 17.1	(p) 88.8		
(a) 35.7	(b) 31.3	(c) 27.8	(e) 23.1	(f) 70.0	(g) 2.6	(h) 3.1	(i) 21.2	(j) 72.2	(k) 19.7	(l) 74.0	(m) 18.3	(n) 75.7	(o) 16.4	(p) 77.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.4	39.2	39.8	42.4	45.3	50.3	54.9	59.0	59.3	55.9	50.1	43.8	40.0		
	DBStd	8.70	4.57	5.08	5.13	5.18	4.97	4.66	4.46	4.33	4.27	4.71	4.81	5.23		
	HDD50	1648	336	288	241	157	57	8	0	0	2	55	194	310		
	HDD65	6085	801	706	701	590	456	304	194	185	276	461	637	774		
	CDD50	1060	0	2	5	18	66	156	279	288	179	59	7	1		
	CDD65	20	0	0	0	0	0	2	7	8	3	0	0	0		
	CDH74	80	0	0	0	0	1	12	33	25	8	1	0	0		
	CDH80	6	0	0	0	0	0	1	4	1	0	0	0	0		
	WSAvg	13.3	16.3	15.7	14.3	12.3	11.3	10.7	10.1	11.1	12.2	14.4	15.7	15.7		
	PrecAvg	27.8	2.4	2.0	1.6	2.0	1.9	2.5	2.2	2.7	2.2	2.6	3.1	2.6		
	PrecMax	37.2	4.8	3.3	4.3	5.5	4.1	7.7	4.2	5.7	4.6	4.9	5.9	4.5		
	PrecMin	19.8	0.5	0.2	0.1	0.1	0.4	0.4	0.6	0.4	0.6	0.7	1.0	0.9		
	PrecStd	4.3	1.2	0.9	1.1	1.6	0.9	1.6	1.0	1.3	1.2	1.0	1.2	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	56.0	60.0	65.8	70.5	75.7	77.8	77.3	75.2	65.5	58.2	54.2		
		MCWB	50.8	50.8	52.1	55.5	61.1	65.5	64.8	66.1	64.0	60.0	55.8	51.9		
	2%	DB	50.6	52.0	55.3	60.0	64.9	69.8	73.4	72.5	69.2	61.6	55.3	52.2		
		MCWB	48.2	48.5	49.9	52.8	58.0	61.8	63.7	63.9	61.2	57.9	53.3	50.2		
	5%	DB	48.2	49.5	52.5	56.7	61.9	66.2	70.2	69.4	65.6	59.1	53.0	50.1		
		MCWB	46.1	46.5	48.1	50.8	55.3	59.7	61.8	61.7	59.0	56.4	51.1	48.3		
	10%	DB	46.2	47.4	50.2	53.9	58.9	63.3	67.2	66.8	62.9	57.1	50.9	47.9		
		MCWB	44.2	44.6	46.3	48.8	53.5	57.7	60.3	60.1	57.4	54.6	49.2	46.1		
	0.4%	WB	51.4	51.3	53.0	56.6	62.8	68.2	67.3	68.9	65.4	61.1	56.2	52.7		
		MCDB	52.9	54.6	57.2	62.1	68.2	73.3	74.9	72.3	71.9	64.3	57.6	53.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	48.5	49.0	50.8	53.7	58.9	63.3	64.9	65.3	62.5	58.7	53.6	50.5		
		MCDB	50.3	51.6	54.5	58.4	64.2	68.2	71.0	70.7	67.3	60.7	54.9	51.9		
	5%	WB	46.5	46.8	48.6	51.3	56.3	60.4	63.0	63.3	60.2	56.9	51.4	48.4		
		MCDB	47.9	49.1	51.8	55.9	60.7	64.9	68.2	67.8	64.2	58.8	52.6	50.0		
	10%	WB	44.4	44.9	46.7	49.2	54.0	58.4	61.2	61.5	58.3	54.8	49.5	46.2		
		MCDB	45.9	47.1	49.8	53.2	58.1	62.5	65.8	65.2	61.7	56.6	50.7	47.7		
	5% DB	MDBR	7.3	8.4	10.4	11.3	11.7	12.2	12.7	12.7	11.5	9.4	8.1	7.5		
		MCDBR	9.9	10.7	13.5	16.2	16.7	17.4	17.7	16.5	15.9	11.5	9.9	9.7		
		MCWBR	9.6	8.9	9.7	10.7	11.3	11.5	9.8	9.7	9.7	9.0	9.2	9.4		
		MCWBR	9.8	10.4	12.8	15.2	15.7	16.6	15.8	14.7	14.3	11.4	9.5	9.8		
	5% WB	MCWBR	9.7	9.1	10.0	10.7	11.4	12.1	10.1	10.2	9.7	9.6	9.1	9.5		
		taub	0.314	0.331	0.360	0.385	0.381	0.386	0.387	0.378	0.358	0.341	0.316	0.299		
		taud	2.346	2.411	2.342	2.295	2.358	2.365	2.366	2.406	2.462	2.473	2.402	2.313		
		Ebn at Noon	200	240	258	266	274	273	270	266	257	234	199	178		
		Edh at Noon	17	23	30	36	36	36	36	32	27	22	16	15		
(44) All-Sky Solar Radiation	RadAvg	188	387	735	1171	1503	1523	1465	1212	872	481	234	142			
	RadStd	24	45	96	123	82	129	124	79	78	41	23	14			

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (48 neighbors)	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A

Nomenclature: See separate page